A Complement-Layer Obstruction to Identical-Constant Nested Spherical Designs

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Abstract

Zheng and Zhuang conjecture that for every pair $t_1 < t$ one can nest a spherical t_1-design of cardinality $C_d t_1^d$ in a spherical t-design of cardinality $C_d t^d$, with the same dimension-dependent constant. We disprove this identical-constant statement. The added layer in any nested pair is itself a t_1-design and therefore has $\Omega_d(t_1^d)$ points. For adjacent strengths $t = t_1 + 1$, however, the conjectured cardinalities leave only $O_d(t_1^{d-1})$ new points. The argument also rules out a common asymptotic leading coefficient and forces $\Omega_d(T^{d+1})$ points in a strict every-degree nested chain through strength T. In the positive direction, regular polygons settle the weaker pairwise optimal-order problem on $\mathbb{S}^1$. Status: *candidate full counterexample, likely valid, pending human review.*

1 Source conjecture and its exact convention

The source is R. Zheng and X. Zhuang, *On the existence and estimates of nested spherical designs* [1]. A multiset $Z_K \subset \mathbb{S}^d$ is a spherical s-design if

$$\frac{1}{K}\sum_{z\in Z_K} P(z) = \int_{\mathbb{S}^d} P\, d\mu_d$$

for every polynomial P of degree at most s. The source explicitly allows repeated points. It calls $Y_M \subset Z_K$ a nested pair when Y_M is a t_1-design, Z_K is a t-design, and $t_1 < t$.

The interpretation of "order" matters here. The source defines N to be of order $h(t_1, t, d)$ when

$$N = C_d h(t_1, t, d)$$

for a constant depending only on d; it separately writes $N \sim t^d$ for two-sided comparability. Figure 1 reproduces this definition. The conjecture in Figure 2 asks for the *same* constant in the cardinalities of the lower- and higher-strength designs for all $t_1 < t$.

> We say that N is of order $h(t_1, t, d)$ for some function h depending on t_1, t, d if $N = C_d \cdot h(t_1, t, d)$ for some constant C_d depending only on d but not the others. The notion $N \sim t^d$ indicates equivalence relation $C_1 t^d \le N \le C_2 t^d$ with constants C_1, C_2 possibly depending on d but not t.

Figure 1: The source's exact definition of "of order", from PDF page 6.

5. Discussion on the optimal order

In contrast to the optimal order t^d for a spherical t-design, our result in Corollary 2 shows that an upper bound of the minimal $N + M$ in *arbitrary* nested spherical t-designs with the t_1-design $\mathcal{Y}_M$ of order t_1^d is of order t^{2d+1}. This is, of course, a rough estimate. Moreover, the properties of t_1-designs are not involved. Following the setting in general spherical t-designs where the constant in the orders are identical for all t, it is thus natural to conjecture that the following statement is true.

Conjecture. *For any $t_1, t \in \mathbb{N}$ such that $t_1 < t$, there exist some spherical t_1-designs of order t_1^d such that they can be extended to a spherical t-design of order t^d. Moreover, the constants (not depending on t_1 and t but d) in the orders are* ***identical*** *for any t and t_1 satisfying $t_1 < t$.*

Figure 2: The identical-constant conjecture, from PDF page 12.

2 Proof intuition

Equal weights make nesting unexpectedly rigid. Both the full design and its old part integrate every polynomial of degree at most t_1 exactly. On subtracting the two unnormalized quadrature identities, the new part also integrates those polynomials exactly with equal weights. Thus every nonempty added layer must itself be a t_1-design.

The source quotes the classical lower bound that any t_1-design requires a constant multiple of t_1^d points. Increasing the strength from t_1 to $t_1 + 1$ while preserving one coefficient C_d changes $C_d t^d$ by only a discrete derivative, of order t_1^{d-1}. These two requirements are incompatible. Notice that this argument targets the conjecture's identical-coefficient clause; it does not rule out optimal power laws with different comparison constants.

3 The layer theorem and the counterexample

For a polynomial P, write $\mathcal{I}(P) = \int_{\mathbb{S}^d} P \, d\mu_d$.

Lemma 3.1 (Complement-layer principle)**.** *Let $Y_M \subset Z_K$ be multisets on $\mathbb{S}^d$, with $K > M$. If Y_M is a spherical s-design and Z_K is a spherical r-design for some $r \geq s$, then the multiset difference $X_{K-M} = Z_K \setminus Y_M$ is a spherical s-design. Consequently,*

$$K - M \geq N(d, s), \tag{1}$$

where $N(d, s)$ is the minimum cardinality of a spherical s-design.

Proof. For every polynomial P of degree at most s, exactness of the two designs gives

$$\sum_{z \in Z_K} P(z) = K\mathcal{I}(P), \qquad \sum_{y \in Y_M} P(y) = M\mathcal{I}(P).$$

Subtracting yields

$$\sum_{x \in X_{K-M}} P(x) = (K - M)\mathcal{I}(P),$$

which is exactly the s-design identity for the added layer. The definition of $N(d,s)$ now gives (1). $\square$

The Delsarte–Goethals–Seidel lower bound quoted in the source [2] is

$$L_d(s) = \begin{cases} \binom{d+k}{d} + \binom{d+k-1}{d}, & s = 2k, \\ 2\binom{d+k}{d}, & s = 2k+1, \end{cases} \qquad N(d,s) \geq L_d(s). \tag{2}$$

In particular,

$$L_d(s) \sim \frac{2^{1-d}}{d!}\, s^d. \tag{3}$$

Theorem 3.2 (Negative resolution of the identical-constant conjecture)**.** *For every $d \geq 1$, there is no constant $C_d > 0$ such that for all integers $t_1 < t$ there is a nested pair with*

$$|Y| = C_d t_1^d, \qquad |Z| = C_d t^d.$$

More generally, adjacent nested pairs cannot have a common asymptotic leading coefficient: there are no pairs satisfying

$$|Y_n| = Cn^d + o(n^d), \qquad |Z_n| = C(n+1)^d + o(n^d)$$

for every sufficiently large n.

Proof. Set $t_1 = n$ and $t = n+1$. Lemma 3.1 and (2) would imply

$$C_d\big((n+1)^d - n^d\big) \geq L_d(n). \tag{4}$$

The left side is $O_d(n^{d-1})$, whereas by (3) the right side is $\Omega_d(n^d)$, contradicting (4) for large n.

Under the asymptotic formulation, the difference $|Z_n| - |Y_n|$ is $o(n^d)$, while Lemma 3.1 still makes it at least $L_d(n) = \Omega_d(n^d)$. This is the same contradiction. $\square$

Corollary 3.3 (Strict every-degree chains are one power larger)**.** *Suppose*

$$Z_1 \subsetneq Z_2 \subsetneq \cdots \subsetneq Z_T$$

and Z_s is a spherical s-design. Then

$$|Z_T| \geq |Z_1| + \sum_{s=1}^{T-1} L_d(s) = \Omega_d(T^{d+1}).$$

Proof. Lemma 3.1 makes every nonempty layer $Z_{s+1} \setminus Z_s$ an s-design, so its cardinality is at least $L_d(s)$. Sum the disjoint layer sizes and use $L_d(s) = \Omega_d(s^d)$. $\square$

Remark 3.4. *Geometrically spaced strengths are consistent with the layer bound: for $s_j \asymp q^j$, the sum $\sum_j s_j^d$ is of the same order as its largest term. This explains why the source's fixed-rational-ratio construction does not conflict with Theorem 3.2.*

4 The weaker optimal-order question is true on the circle

The preceding obstruction leaves open whether both sizes can merely be comparable to t_1^d and t^d with nonidentical constants. For $d = 1$ this weaker question has a completely explicit positive answer.

Theorem 4.1. *For every pair of integers $1 \le t_1 < t$, there are nested designs $Y \subset Z$ on $\mathbb{S}^1$ such that*

$$t_1 < |Y| \le 2t_1, \qquad t < |Z| \le 2t.$$

Proof. Identify $\mathbb{S}^1$ with the unit complex numbers. Put

$$q = t_1 + 1, \qquad \ell = \left\lfloor \frac{t}{q} \right\rfloor + 1.$$

Let Y be the regular q-gon and Z the regular $q\ell$-gon, arranged as the corresponding sets of roots of unity. Since q divides $q\ell$, we have $Y \subset Z$.

The regular m-gon is an $(m-1)$-design: its averages of z^k vanish for $0 < |k| < m$, and restrictions of real polynomials of degree at most $m - 1$ are linear combinations of these Fourier modes. Hence Y is a t_1-design. Since $q\ell > t$, Z is a t-design. Finally,

$$q = t_1 + 1 \le 2t_1, \qquad q\ell \le t + q \le 2t,$$

where $q \le t$ follows from $t_1 < t$. □

5 Verification, scope, and novelty status

The complement identity is an exact subtraction of the two equal-weight quadrature formulas. The cardinality contradiction uses the lower bound quoted by the source under its own multiset convention. The circle construction was also checked directly through roots-of-unity power sums.

Theorem 3.2 refutes the identical-coefficient conjecture as formulated using the source's definition of "order". It does not refute the weaker higher-dimensional possibility $|Y| \asymp_d t_1^d$, $|Z| \asymp_d t^d$ with unrelated constants. Nor does Corollary 3.3 apply to a chain that reuses the same already higher-strength design at several levels; it concerns strict new layers at every consecutive strength.

A bounded novelty search on 11 August 2026 used the exact title, arXiv id, conjecture wording, and complement/layer/identical-constant keywords. The January 2026 survey of An and Zhuang still describes the $O(t^d)$ optimal bound as believed and reports only the fixed-ratio case [3]. No direct complement-layer obstruction, adjacent-strength counterexample, or circle construction above was found. This was not an exhaustive literature review.

Human review should focus on the intended force of "identical constants" and on the source's use of the Delsarte lower bound under its explicit multiset convention.

References

[1] R. Zheng and X. Zhuang, *On the existence and estimates of nested spherical designs*, Appl. Comput. Harmon. Anal. **73** (2024), 101672, doi:10.1016/j.acha.2024.101672; arXiv:2405.10607.

[2] P. Delsarte, J.-M. Goethals, and J. J. Seidel, *Spherical codes and designs*, in *Geometry and Combinatorics*, Elsevier, 1991, pp. 68–93.

[3] C. An and X. Zhuang, *A survey on spherical designs: existence, numerical constructions, and applications*, arXiv:2601.11963 (2026).